

- 19 Study the number pattern shown in the table below:

Number (n)	Pattern	Sum of Number Pattern (S)	$S + 2$
1	2	2	4
2	2, 3, 2	7	9
3	2, 3, 4, 3, 2	14	16
4			

- (a) Complete the table. [1]
- (b) If $S = 2 + 3 + 4 + \dots + (k - 1) + k + (k + 1) + k + (k - 1) + \dots + 4 + 3 + 2 = 98$, find the value of k .

Answer $k =$ [1]

- (c) Write a formula connecting S and n .

Answer [1]

- (d) Hence, find the sum of 2, 3, 4, ..., 98, 99, 98, ..., 4, 3, 2.

Answer [1]

- 20 In a quadratic equation, the coefficients of x and x^2 are m and 5 respectively.
The constant of the quadratic equation is -16 .
One solution of the quadratic equation is $-\frac{2}{5}$.

Find m and the other solution of the equation.

Answer $m =$

$x =$ [3]

21 $\frac{3a-bc}{2ac-5b} = \frac{1}{2}$

- (a) Rearrange the formula to make c the subject.

Answer $c =$ [3]

- (b) Calculate the value of c when $a = -3$ and $b = 2$.

Answer [1]